MIL-B-43714

4 September 1970

SUPERSEDING
(see 6.3)

MILITARY SPECIFICATION

BALANCES, TRIP, LABORATORY

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope.- This specification covers laboratory trip balances.

1.2 Classification.- The balances shall be of the following styles and sizes, as specified (see 6.2).

Style 1 - Single beam, 5000 gram capacity
Style 2 - Double beam

Size 1 - 8 pound, 9 ounce capacity
Size 2 - 2000 gram capacity

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATION

FEDERAL

PPP-B-1122 - Balances, Scales, and Accessories, Packaging and Packing of.

STANDARD

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

FSC 6670

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(Copies of specifications and standards required by suppliers in connnection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publication.- The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS (NBS)

NBS Circular 547 Section 1 - Precision Laboratory Standards of Mass And Laboratory Weights

(Application for copies of Circular 547 should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal Agencies.)

3. REQUIREMENTS

3.1 First article.- This specification contains provisions for first article test and approval (see 4.2, 6.2, and 6.4).

3.2 Standard product.- The balances covered by this specification shall be the current standard product of the manufacturer, except for changes necessary to comply with this specification.

3.3 Material.- Material not definitely specified shall be of the quality normally used by the manufacturer for trip balances, provided the completed item complies with all the provisions of this specification.

3.4 Design.- The balances shall be equal arm scales. The base of the balance shall be cast metal. The bottom of the base shall be provided with a hole not less than 1/2 inch in diameter and not less than the equivalent of one diameter in depth for insertion of a rod to elevate the balance above a resting surface. Each balance shall have a balance indicator consisting of a pointer extending to a graduated scale. The graduated scale shall have a center zero and not less than four graduations, to the left and right of the zero marking.

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3.4.1 Weighbeams.- The style 1 balance shall have a single weighbeam. The style 2 balance shall have a double weighbeam. Weighbeams shall be fabricated of either corrosion-resisting steel or aluminum alloy and shall be graduated as specified in table I. Weighbeams shall be provided with sliding poises. Weighbeams of 8 ounce and 200 gram capacity shall be notched. Other weighbeams shall be smooth.

TABLE I.- Weighbeams

	First Beam		Second Beam	
	Capacity	Graduation	Capacity	Graduation
Style 1	10 grams	0.1 gram	--	--
Style 2 - Size 1	1 ounce	1/64 ounce	8 ounces	1/8 ounce
Style 2 - Size 2	10 grams	0.1 gram	200 grams	10 grams

3.4.2 Load receivers.- Unless otherwise specified (see 6.2), the balances shall be provided with the load receivers specified in table II.

TABLE II.- Load receivers

	Number Required	Material	Type	Diameter, Minimum
Style 1	2	Ceramic	Plate	5-1/2 inches
Style 2 - Size 1	2	Ceramic	Plate	5-1/2 inches
Style 2 - Size 2	2	Corrosion-Resisting Steel	Plate	6 inches

3.4.3 Accessories.-

3.4.3.1 Covers.- Each balance shall be provided with a cover. The cover shall be of sufficient size to completely cover the balance.

3.4.3.2 Stand.- When specified (see 6.2), the balance shall be provided with a stand consisting of an adjustable clamp and support rod which fits into the hole in the base of the balance.

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3.4.3.3 Weights.- When specified (see 6.2), weight sets shall be furnished with the balances. Individual weights and weight sets shall conform to class T of National Bureau of Standards (NBS) Circular No. 547. The weight set furnished with the style 1 balance shall provide for a weighing capacity to 5000 grams. The weight set for the style 2, size 2 balance shall consist of one 1000-gram, one 500-gram and two 200-gram weights. The style 2, size 1 balance shall be furnished with a weight set consisting of the following:

Weight (ounces)	Quantity (each)
64	1
32	1
16	1
8	1
4	1
2	1
1	1
1/2	1
1/4	1
1/8	1
1/16	2

3.5 Performance requirements.-

3.5.1 Overload.- The load receivers loaded to 125 percent of the balance capacity shall show no evidence of deformation or misalignment of bearings and knife edges when tested as specified in 4.4.1.

3.5.2 Sensitivity requirement.- When tested as specified in 4.4.2, at zero load and at full load, the sensitivity shall be as follows:

Style	Zero load sensitivity grams per graduation	Full load sensitivity grams per graduation
1	0.5 gram	1.0 gram
2 Size 1	0.5 gram	1.0 gram
2 Size 2	0.1 gram	0.2 gram

3.6 Finish.- The balance shall be finished in accordance with the manufacturer's commercial finish.

3.7 Marking for identification.- The manufacturer shall permanently and legibly mark the capacity of balance and his name or trade mark of such known characters identifiable with said manufacturer on each balance.

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3.8 Operating instructions.- Operating instructions in accordance with the manufacturer's commercial practice shall be furnished with each balance.

3.9 Workmanship.- The balances shall be free of fractures, burrs, or exposed sharp edges. Enameled surfaces shall be free of runs, wrinkles, voids, or imbedded foreign matter. Dust covers shall be free of cracks and tears.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection.- When a preproduction sample is required, it shall be examined for the defects in table III, specified dimensions, and tested as specified in 4.4.1 and 4.4.2. The presence of any visual defect, any dimension not within the specified requirements, or failure to pass the tests shall be cause for disapproval of the preproduction sample.

4.3 Inspection.- Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.3.1 Component and material inspection.- In accordance with 4.1, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification, or applicable purchase document.

4.3.2 End item inspection.- A lot shall consist of all balances of one style and size offered for inspection at one time. The sample unit for this inspection shall be one complete balance.

4.3.2.1 Visual examination.- Examination shall be made of the balance for defects set forth in table III. The inspection level shall be level II with an acceptable quality level (AQL) of 2.5 for major defects and 6.5 for total defects, expressed in terms of defects per hundred units.

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TABLE III.- Visual examination of defects

Examine	Defect	Classification Major	Classification Minor
Finish	Run, wrinkle, void, or foreign matter imbedded in enameled surface.		X
Design and workmanship	Not style specified.	X	
	Hole missing from base bottom or not as specified.	X	
	Burr, fracture, or exposed sharp edge.	X	
Weighbeams	Not graduated as specified. Sliding poise missing.	X	
	8 ounce and 200 gram weighbeams, as applicable, not notched.		X
	Other weighbeams not smooth.		X
Load receivers	Missing, not material specified, not type specified.	X	
Accessories:			
Dust cover	Missing, not specified size, tear, or crack.		X
Stand (when specified)	Missing, rod does not fit balance hole, component missing.	X	
Weights or weight sets (when specified)	Weight missing or not as specified.	X	
Balance	Pointer missing.	X	
Indicator	Graduated scale or zero balance missing.	X	
	Number of graduations on scale less than specified.		X
Marking	Missing, incomplete, not legible.		X
Operating instructions	Missing, incomplete, not legible.		X

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4.3.2.2 Dimensional examination.- Examination shall be made of the balance for compliance with dimensions specified. Any deviation from specified requirements shall constitute a defect. The inspection level shall be S-2 with an AQL of 4.0, expressed in terms of defects per hundred units.

4.3.2.3 Testing of the end item.- The end item shall be tested as specified in 4.4.1 and 4.4.2. The inspection level shall be S-2 with an AQL of 2.5 expressed in terms of defects per hundred units.

4.3.3 Examination of preparation for delivery.- Examination of preparation for delivery shall be in accordance with PPP-B-1122.

4.4 Tests.-

4.4.1 Overload test.- A load equal to 125 percent of the rated load of the balance shall be placed in each load receiver. The load shall then be removed. Any evidence of deformation or misalignment of the bearings and knife edges shall constitute failure of this test.

4.4.2 Sensitivity test.- At zero load, for style 1 and style 2, size 1 balances, a 0.5 gram load shall be applied to one load receiver. At zero load for the style 2, size 2 balance, a 0.1 gram load shall be applied to one load receiver. Weights shall be applied for not less than 5 minutes for the test. Failure of the pointer to move to an imbalanced condition constitutes failure of this test. The test shall be repeated at full load, except a 1 gram weight shall be removed from the load receiver of the style 1 and style 2, size 1 balance and a 0.2 gram weight shall be removed from one load receiver of the style 2, size 2 balance.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing, and marking for level A, B, or C shall be in accordance with PPP-B-1122 (see 6.2).

6. NOTES

6.1 Intended use.- The balances covered by this specification are intended for general laboratory use. The balance is used with a stand for making specific gravity determinations.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Style and size required (see 1.2).
(c) First article (see 3.1, 4.2, and 6.4).
(d) When load receivers are other than specified (see 3.4.2).

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(e) When stand is required (see 3.4.3.2).
(f) When weights or weight sets are required (see 3.4.3.3).
(g) Selection of applicable levels of preservation, packaging, and packing (see 5.1).

6.2.1 Any requirements for equipment manuals for the item covered by this specification should be included in DD Form 1423 Contract Data Requirements List and cited in the contract.

6.3 Supersession data.- This specification includes the requirements of Defense Medical Purchase Description Number 7, dated 7 December 1966.

6.4 First article.- The item covered by this specification requires first article test and approval under the appropriate provisions of paragraph 7-104.55 of the Armed Forces Procurement Regulations. The first article should be a preproduction sample consisting of one unit. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for examination, test, and approval of the first article.

Custodians:

Army - GL
Air Force - 82

Review activities:

Army - ME, MU
Air Force - 03

User activities:

Navy - YD, MS

Preparing activity:

Army - GL

Project No. 6670-0106

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION
Balances, Trip, Laboratory MIL-B-43714

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
   ☐ YES ☐ NO *(If "yes", in what way?)*

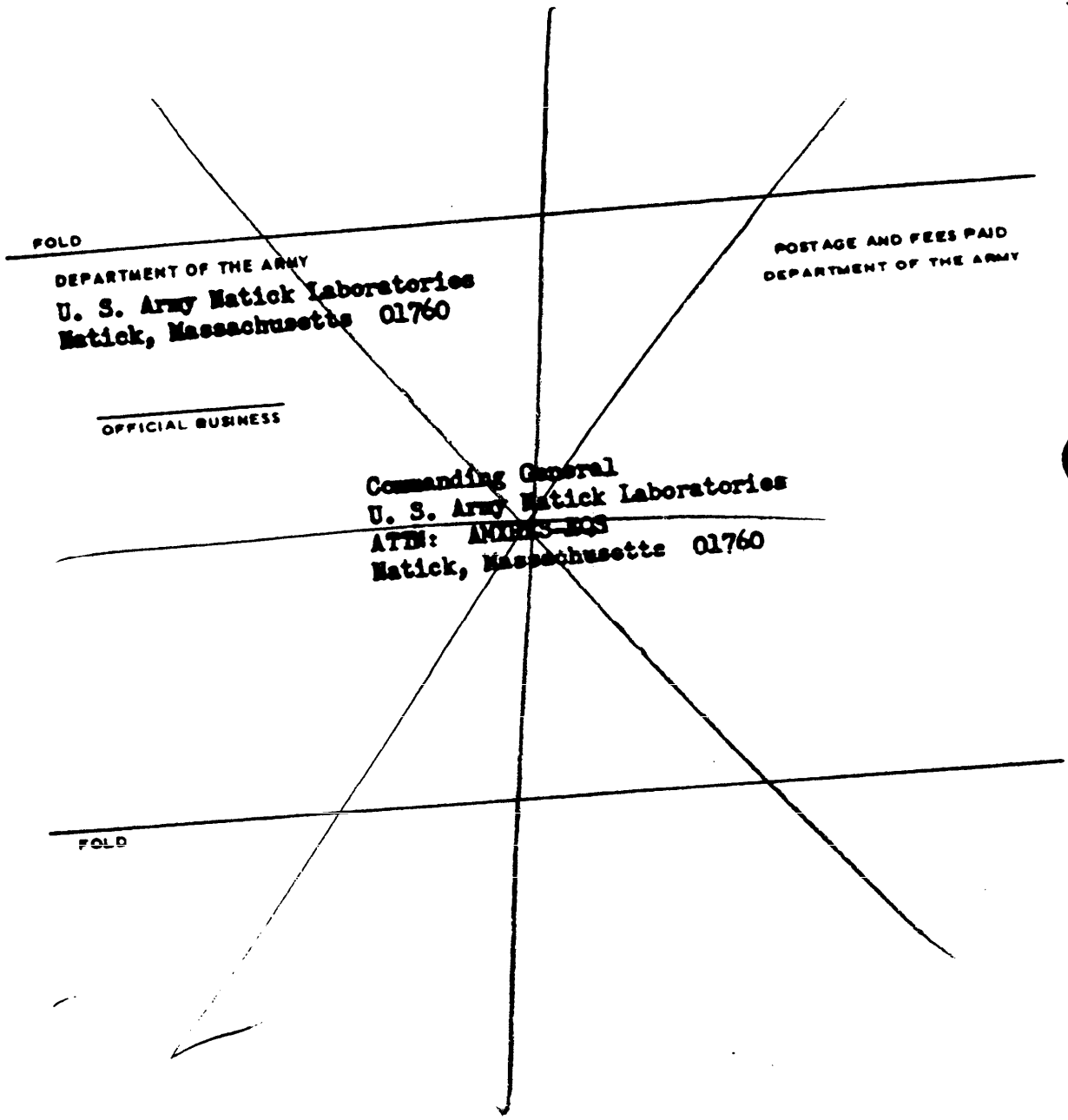
4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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